Einstein's General Theory of Relativity

Lecture 12: The Schwarzschild Horizon and Black-Hole Interior

Lectures by Leonard Susskind

Transcript-derived companion notes curated by LazyingArt LLC
Produced with Video2Book

Chapter 1

Lecture 12: The Schwarzschild Horizon and Black-Hole Interior

Overview. The geometry near a Schwarzschild horizon is first recognized in a simpler setting: flat spacetime described by uniformly accelerated observers. Hyperbolic coordinates produce the Rindler metric, an observer-dependent horizon, and the asymmetry between an inertial observer who crosses that horizon and an accelerated observer who remains outside. The Schwarzschild metric appears to be singular at $r = 2GM$, but a proper-distance coordinate and a rescaled time reduce its near-horizon form exactly to the Rindler metric. The horizon is therefore not a curvature singularity. It is nevertheless a causal boundary for observers who stay outside. The true Schwarzschild singularity is at $r = 0$, where tidal curvature diverges and which lies as a spacelike future boundary inside the hole. The closing discussion returns to the vacuum Einstein equations, the Newtonian exterior problem, gravitational collapse, mean-density scaling, cosmic horizons, and the limits of classical physics at the singularity.

We use signature $+---$, set $c = 1$, and write

$$r_{\mathrm{s}} = 2GM \tag{1.1}$$

for the Schwarzschild radius. The symbol ρ will denote proper distance from a horizon; r is reserved for the Schwarzschild areal radius.

1.1 The Accelerated Congruence

Before entering a black hole, return to flat spacetime. In an ordinary inertial diagram, every observer in the accelerated congruence follows a hyperbola. The curves nearer the null boundary turn through a large rapidity in a shorter proper distance and therefore require greater proper acceleration. This differs from the Newtonian picture in which a rigid array is assigned one common acceleration. Relativity permits a stationary accelerated array only when its proper acceleration varies from one worldline to the next.

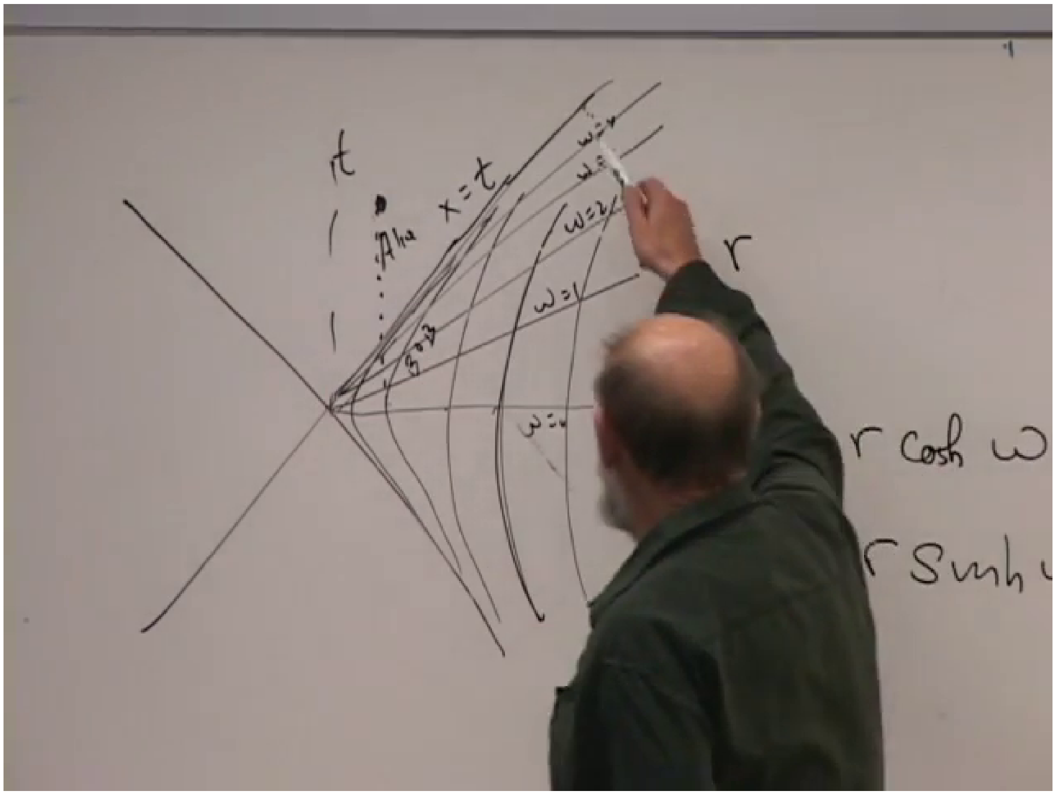

Figure 1.1: The Rindler congruence, constant-ω rays, and the coordinate map $X = \rho \cosh \omega$, $T = \rho \sinh \omega$.

Lecture frame: 00:09:55.

Question from the audience. Does this show that special relativity cannot be used for accelerated observers? [a]

Response. No. The entire construction takes place in flat Minkowski spacetime and is governed by special relativity. Each fixed-position observer is accelerated, and different positions require different accelerations. That variation is also what one expects for observers supported at different heights in a gravitational field.

[a]Lecture timestamp: 00:01:35–00:02:54.

Introduce inertial coordinates (T, X, y, z) and hyperbolic coordinates (ω, ρ, y, z):

$$X = \rho \cosh \omega, \qquad T = \rho \sinh \omega. \tag{1.2}$$

These equations are the Lorentzian counterparts of $x = \rho \cos \theta$, $y = \rho \sin \theta$. They imply

$$X^2 - T^2 = \rho^2, \qquad \tanh \omega = \frac{T}{X}. \tag{1.3}$$

Thus $\rho =$ constant labels hyperbolic worldlines and $\omega =$ constant labels straight rays through the origin. The rapidity ω ranges from $-\infty$ to $+\infty$.

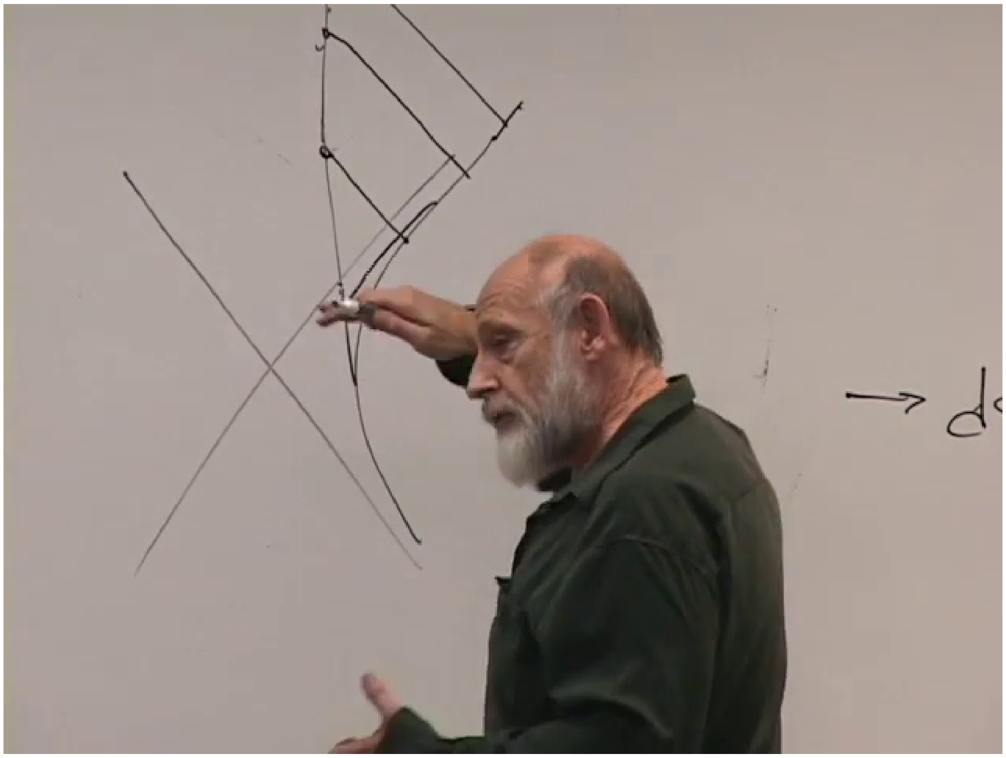

Figure 1.2: Past-directed null rays used by Bob to see Alice. Successive reception events intersect Alice's inertial worldline closer to the Rindler horizon.

Lecture frame: 00:12:55.

1.1.1 What Alice and Bob See

The null lines $X = \pm T$ divide Minkowski spacetime into wedges. An observer confined to the right Rindler wedge cannot receive a signal emitted beyond its past or future null boundary. The future boundary $X = T$ is approached as $\omega \to \infty$. It is an acceleration horizon, not a black-hole horizon: there is no curvature and no gravitating body.

Let Alice release herself and cross the null boundary inertially. Let Bob remain on a fixed-ρ hyperbola. Seeing means receiving light, so Bob locates Alice by tracing past-directed null rays from successive events on his worldline. Each later ray meets Alice closer to the horizon. Bob therefore sees her approach the boundary asymptotically. Alice crosses it after finite proper time.

Bob sees Alice's processes slow and her longitudinal dimensions Lorentz contract as their relative rapidity grows. Alice does not see a wall or a singular event. Looking backward, she sees Bob accelerating away. His signals are also Doppler shifted and slowed, but he does not disappear behind a horizon in her causal past.

Question from the audience. Does Bob see Alice's time slow, and what does Alice see when she looks back at Bob? [a]

Response. Bob receives Alice's successive signals farther apart and says that her clock slows as she approaches his horizon. Alice sees Bob receding on an accelerated trajectory. She can continue to receive signals from him; she does not assign him the same one-way disappearance that he assigns to her. The asymmetry comes from Bob's acceleration and from the different null rays available to the two observers.

[a] Lecture timestamp: 00:11:04–00:12:59.

1.1.2 Clock Rates in the Accelerated Frame

Substitution into the Minkowski line element gives

$$\begin{aligned} ds^2 &= dT^2 - dX^2 - dy^2 - dz^2 \\ &= \rho^2 d\omega^2 - d\rho^2 - dy^2 - dz^2. \end{aligned} \tag{1.4}$$

For an observer at fixed ρ, y, z,

$$d\tau = \rho\, d\omega. \tag{1.5}$$

The coordinate ω is a common time label, not a common proper time. Identical proper-time clocks at different ρ advance at different rates per unit ω. Conversely, clocks engineered to display ω must be calibrated differently at different positions.

The proper acceleration of a fixed-ρ observer is

$$a(\rho) = \frac{1}{\rho}. \tag{1.6}$$

The acceleration diverges as the observer approaches $\rho = 0$, the Rindler horizon.

Question from the audience. If the transformation is only a change of coordinates, where does the acceleration enter? [a]

Response. The map itself is only a coordinate transformation. The physical statement comes when observers are placed on $\rho =$ constant curves. Those curves are hyperbolae in the inertial diagram, so the observers occupying them accelerate. A coordinate chart does not accelerate anyone; a chosen congruence of worldlines does.

[a] Lecture timestamp: 00:13:02–00:13:57.

Question from the audience. Is $d\omega$ the proper-time increment on every fixed-ρ worldline? [a]

Response. No. Equation (1.5) gives $d\tau = \rho\, d\omega$. The factor of ρ is constant along one worldline but differs between worldlines. A common coordinate time can be imposed, but it is not the reading of identical local clocks everywhere.

[a] Lecture timestamp: 00:13:59–00:15:30.

1.1.3 Equal acceleration is not a rigid accelerated frame

A translated copy of one hyperbola has the same proper acceleration as the original. It does not, however, belong to the stationary Rindler array. Equal coordinate separation in the original

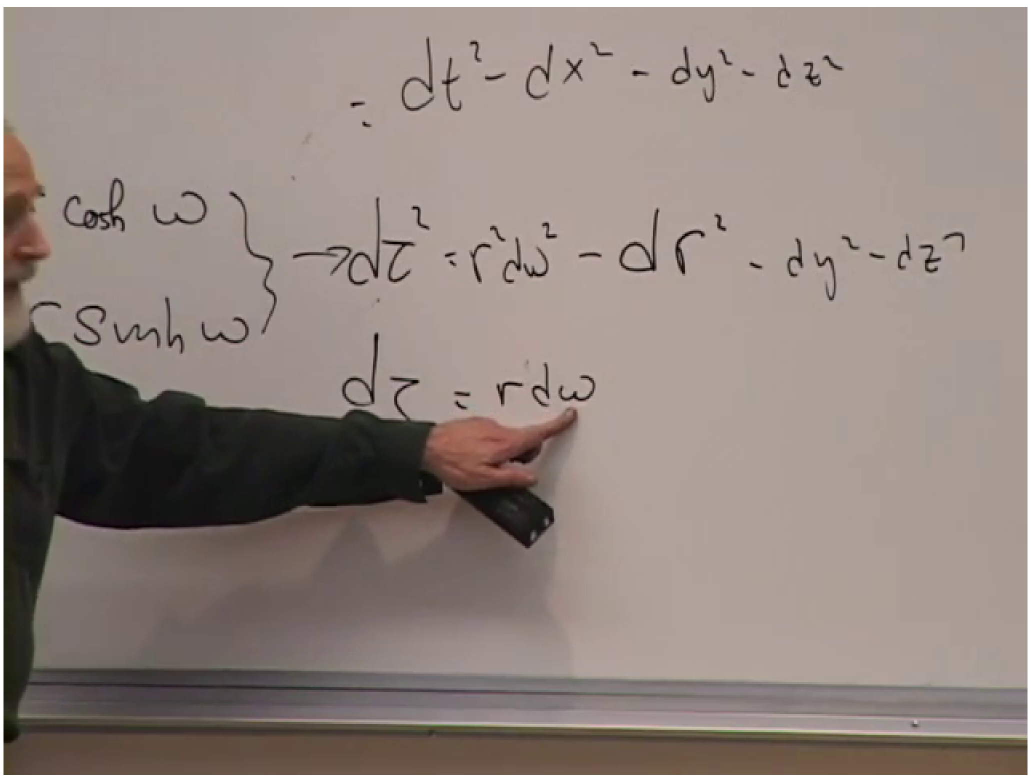

Figure 1.3: The board relation $d\tau = \rho\, d\omega$, showing why equal increments of accelerated time do not represent equal proper time at different ρ.

Lecture frame: 00:14:50.

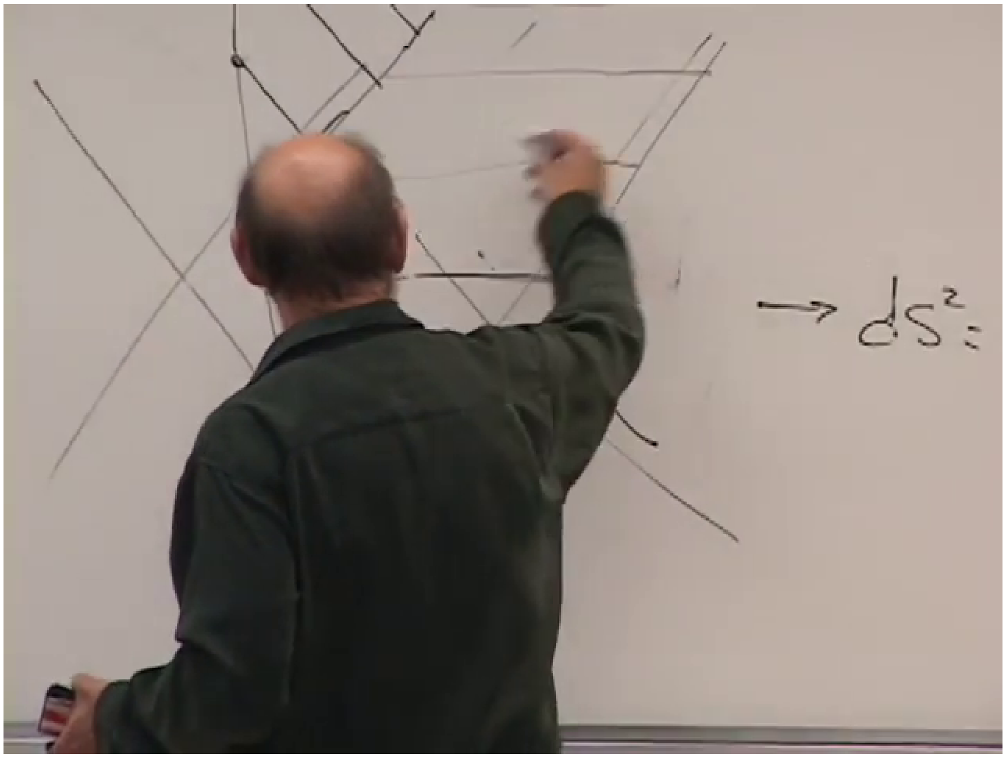

Figure 1.4: Translated equal-acceleration hyperbolae compared with the stationary Rindler congruence. The inertial separations may match while the observers' instantaneous rest-frame separations do not.

Lecture frame: 00:18:55.

inertial frame is not equal separation on the instantaneous rest slices of the accelerated observers. Lorentz contraction changes the latter as the motion develops. The fixed-ρ family instead changes acceleration with position so that neighboring observers retain a constant radar distance.

Question from the audience. Would two translated hyperbolae of the same shape have different gravitational potentials or forces? [a]

Response. The observers have the same proper acceleration and feel the same support force. An additive constant in a Newtonian potential is not observable. What fails is rigidity: translated hyperbolae do not remain at fixed mutual distance when distance is measured on their changing rest slices. The Rindler congruence uses different accelerations precisely to avoid that time-dependent separation.

[a] Lecture timestamp: 00:15:51–00:19:50.

1.2 The Schwarzschild Exterior

The accelerated-frame picture becomes useful because the Schwarzschild horizon has the same local geometry. In Schwarzschild coordinates,

$$ds^2 = \left(1 - \frac{2GM}{r}\right) dt^2 - \frac{dr^2}{1 - \frac{2GM}{r}} - r^2 d\Omega^2, \qquad d\Omega^2 = d\theta^2 + \sin^2\theta\, d\phi^2. \tag{1.7}$$

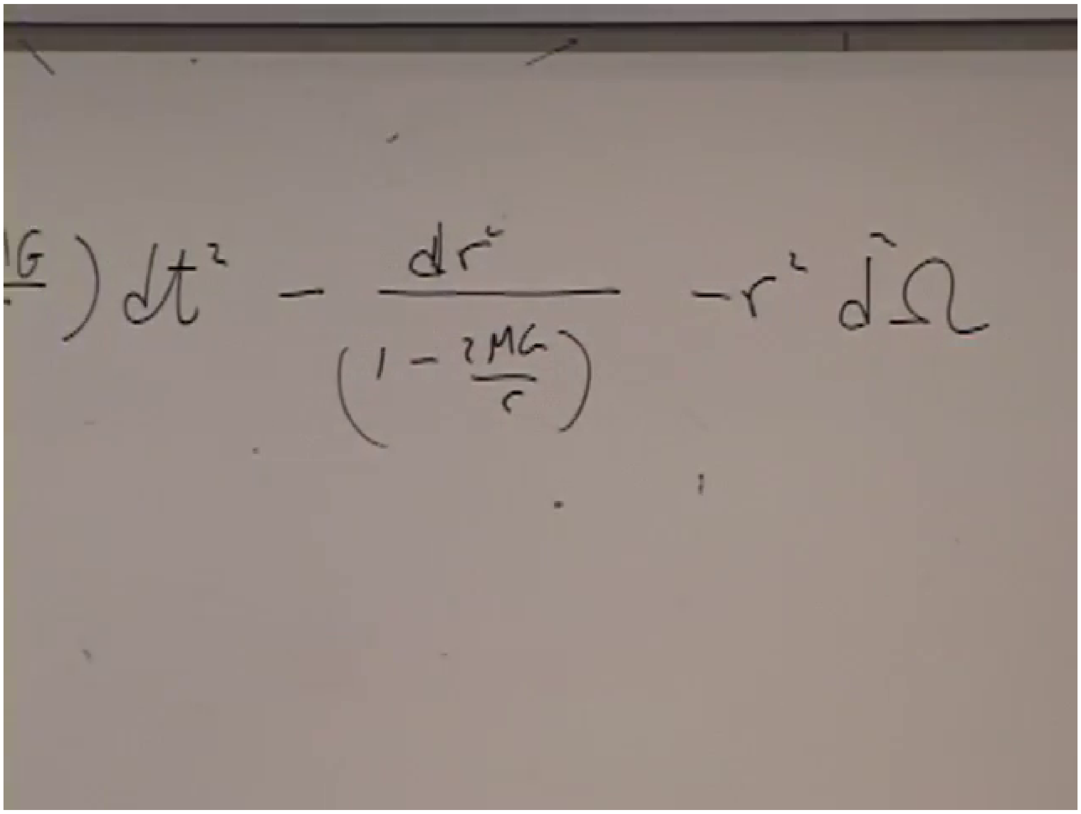

Figure 1.5: The Schwarzschild line element on the board. The complete normalized equation is given in (1.7).

Lecture frame: 00:22:30.

This is the exterior metric of every spherically symmetric body. It describes the field outside the Earth or a star as well as the vacuum region outside a black hole. For an extended body it need not hold through the matter; after complete collapse the vacuum solution extends down to the singular region.

Two radii look suspicious. At $r = r_s$, the coefficient of dt^2 vanishes while the coefficient of dr^2 diverges. At $r = 0$, the coefficients also diverge or vanish and their signs are reversed relative to the exterior. A divergent metric component does not by itself prove a physical singularity. We must ask whether an invariant curvature or a freely falling measurement diverges.

The distinction is decisive:

$$\begin{aligned} R_{\mu\nu} &= 0, \qquad\qquad R = 0 \qquad\qquad \text{for } r > 0, \\ R_{\alpha\beta\gamma\delta}R^{\alpha\beta\gamma\delta} &= \frac{48G^2M^2}{r^6}. \end{aligned} \tag{1.8}$$

The Kretschmann scalar is finite at $r = 2GM$ and diverges at $r = 0$. The horizon singularity is a coordinate failure; the central singularity is curvature.

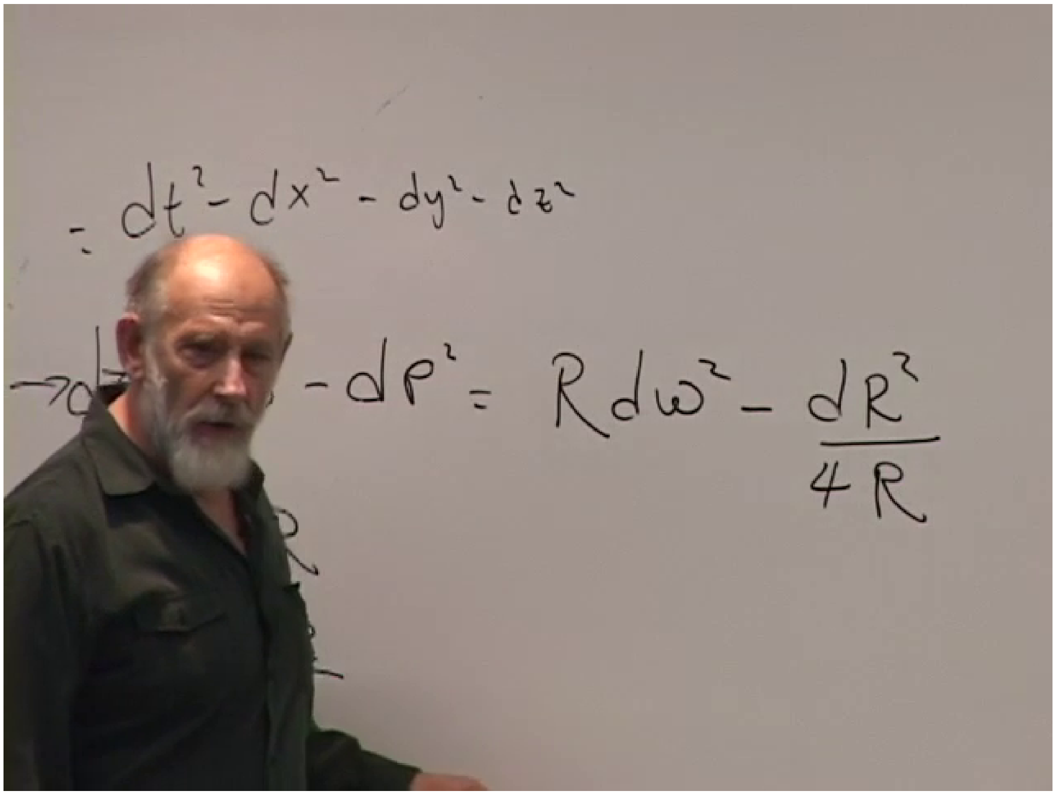

Figure 1.6: The flat Rindler metric rewritten with $R_{\text{aux}} = \rho^2$. Its zero and pole at $R_{\text{aux}} = 0$ imitate the Schwarzschild coordinate singularity without introducing curvature.

Lecture frame: 00:32:40.

1.3 A Coordinate Singularity Manufactured in Flat Space

The Rindler metric can be made to look as alarming as the Schwarzschild metric. Define an auxiliary coordinate

$$R_{\text{aux}} = \rho^2. \tag{1.9}$$

Then

$$d\rho^2 = \frac{dR_{\text{aux}}^2}{4R_{\text{aux}}}, \tag{1.10}$$

and flat spacetime becomes

$$ds^2 = R_{\text{aux}}\, d\omega^2 - \frac{dR_{\text{aux}}^2}{4R_{\text{aux}}} - dy^2 - dz^2. \tag{1.11}$$

At $R_{\text{aux}} = 0$, one coefficient vanishes and another diverges. No curvature has appeared. The singularity was created entirely by replacing ρ with ρ^2.

The apparent exchange of space and time is equally harmless in this example. Since

$$X^2 - T^2 = R_{\text{aux}}, \tag{1.12}$$

positive R_{aux} labels right-wedge timelike hyperbolae. For negative R_{aux},

$$T^2 - X^2 = -R_{\text{aux}}, \tag{1.13}$$

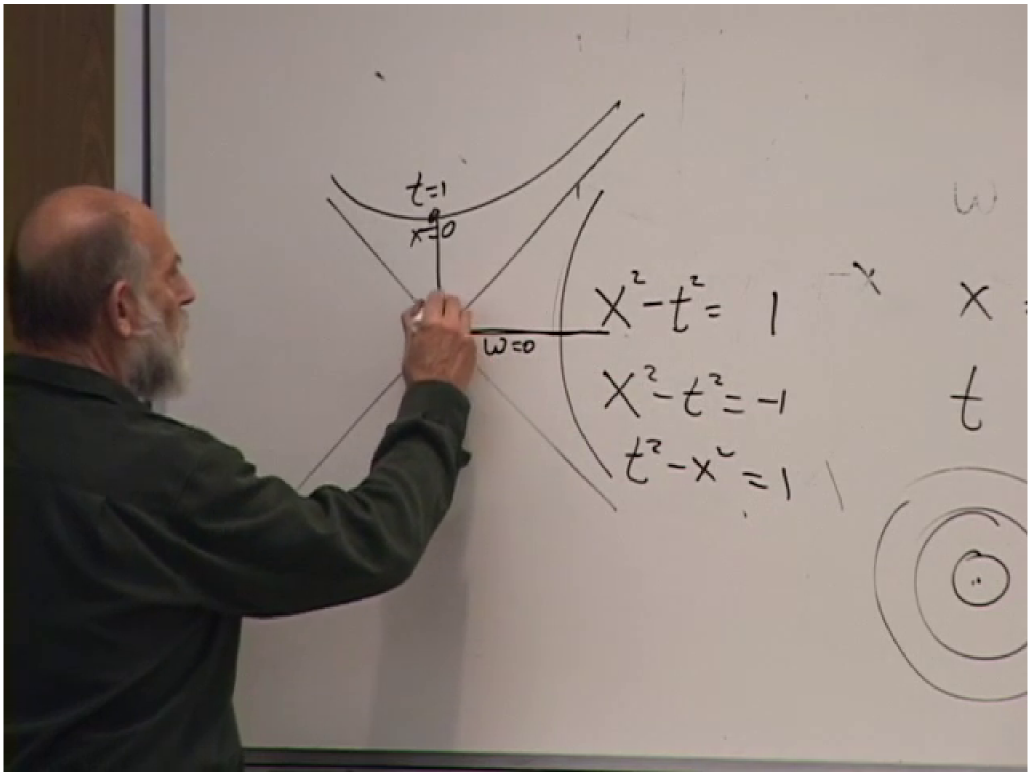

Figure 1.7: Constant-R_{aux} hyperbolae on opposite sides of the Rindler horizon. The sign change turns fixed-coordinate curves from timelike to spacelike without changing the underlying flat geometry.

Lecture frame: 00:37:55.

and the same coordinate labels upper-wedge spacelike hyperbolae. Nothing rotates the physical axes. The coordinate grid changes character across its null boundary.

Question from the audience. How can ρ^2 become negative? [a]

Response. The coordinate ρ was defined only in the right wedge, so its square is nonnegative there. The analytic continuation is made with $R_{\mathrm{aux}} = X^2 - T^2$. That quantity is negative in the future wedge, where the level sets are spacelike hyperbolae. It is the continued coordinate, not a real square of a real right-wedge distance, that changes sign.

[a] Lecture timestamp: 00:34:25–00:38:30.

1.4 The Near-Horizon Approximation

Write

$$1 - \frac{r_{\mathrm{s}}}{r} = \frac{r - r_{\mathrm{s}}}{r}. \tag{1.14}$$

We examine a small neighborhood of one point on the horizon. The difference $r - r_{\mathrm{s}}$ must be retained because it is small, vanishes at the horizon, and changes sign across it. Factors of r that vary only by a tiny fractional amount may be replaced by r_{s}. Therefore

$$ds^2 \simeq \frac{r - r_{\mathrm{s}}}{r_{\mathrm{s}}}\, dt^2 - \frac{r_{\mathrm{s}}}{r - r_{\mathrm{s}}}\, dr^2 - r_{\mathrm{s}}^2 d\Omega^2. \tag{1.15}$$

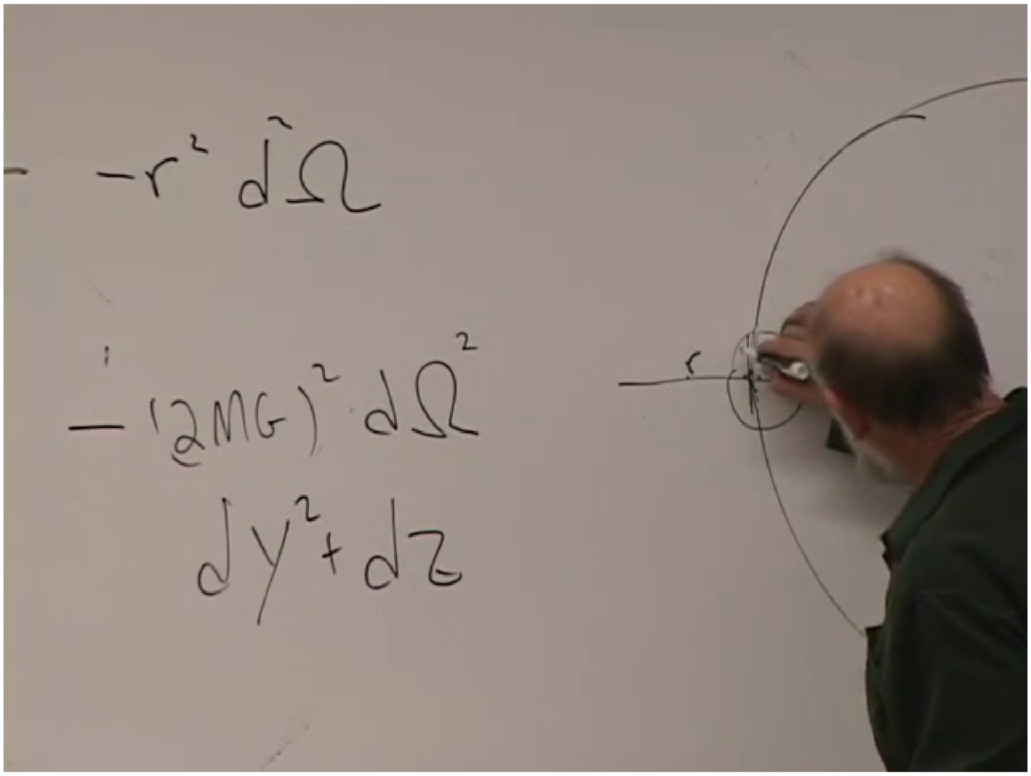

Figure 1.8: A small patch of the horizon two-sphere replaced by its tangent plane, so $r_s^2 d\Omega^2 \simeq dy^2 + dz^2$.

Lecture frame: 00:45:10.

The angular part is a sphere of radius r_s. Under a microscope, a small patch of that sphere is its tangent plane. With local Cartesian coordinates y, z,

$$r_s^2 d\Omega^2 \simeq dy^2 + dz^2, \tag{1.16}$$

and hence

$$ds^2 \simeq \frac{r - r_s}{r_s}\, dt^2 - \frac{r_s}{r - r_s}\, dr^2 - dy^2 - dz^2. \tag{1.17}$$

Question from the audience. Is the useful approximation determined by how far one is from the horizon? [a]

Response. Yes, and also by how small a patch of the horizon is examined. Near $r = r_s$, slowly varying factors may be evaluated at r_s, but the difference $r - r_s$ must remain. On a sufficiently small angular patch the horizon sphere is also well approximated by a plane.

[a] Lecture timestamp: 00:38:36–00:41:55.

1.4.1 Proper distance from the horizon

At fixed t, y, z, define ρ as radial proper distance from the horizon:

$$d\rho = \sqrt{\frac{r_s}{r - r_s}}\, dr. \tag{1.18}$$

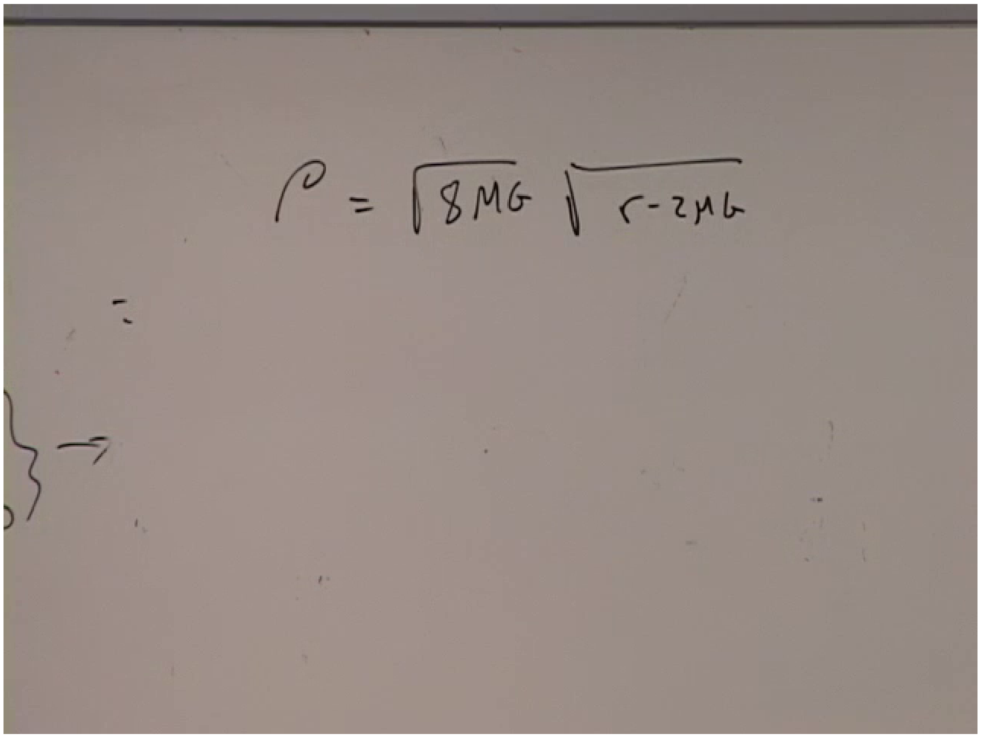

Figure 1.9: The completed proper-distance relation $\rho = \sqrt{8GM(r-2GM)}$.
Lecture frame: 00:56:25.

Integrating from r_s to an exterior point r,

$$\begin{aligned}\rho(r) &= \int_{r_s}^{r} \sqrt{\frac{r_s}{r'-r_s}}\,dr' \\ &= 2\sqrt{r_s(r-r_s)} = \sqrt{8GM(r-2GM)}.\end{aligned} \tag{1.19}$$

Equivalently,

$$r - r_s = \frac{\rho^2}{4r_s} = \frac{\rho^2}{8GM}. \tag{1.20}$$

Question from the audience. Why are the other coordinate increments set to zero, and what is the upper limit of the proper-distance integral? [a]

Response. We want the distance measured along a radial curve on a constant-time slice, so $dt = dy = dz = 0$. The lower endpoint is the horizon $r' = r_s$; the upper endpoint is the exterior point whose Schwarzschild coordinate is r. A primed r' is used only as the integration variable. Taking the square root of the radial metric coefficient supplies the factor $\sqrt{r_s}$ that must remain in the result.

[a] Lecture timestamp: 00:48:44–00:53:57.

Use (1.20) in (1.17). The radial term is $-d\rho^2$ by construction, and the time term becomes

$$\frac{r-r_s}{r_s}dt^2 = \frac{\rho^2}{4r_s^2}dt^2. \tag{1.21}$$

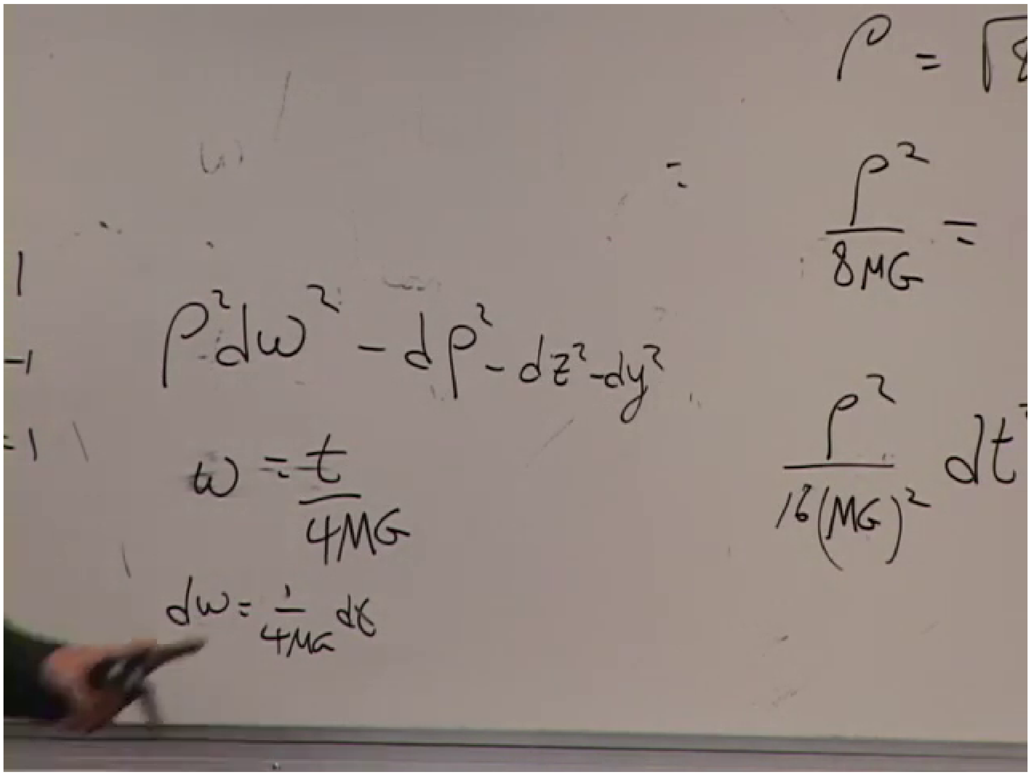

Figure 1.10: The final rescaling $\omega = t/(4GM)$ and the near-horizon Rindler metric.

Lecture frame: 01:00:50.

Now define the dimensionless accelerated time

$$\omega = \frac{t}{2r_s} = \frac{t}{4GM}, \qquad d\omega = \frac{dt}{4GM}. \tag{1.22}$$

The metric becomes

$$\boxed{ds^2 \simeq \rho^2 d\omega^2 - d\rho^2 - dy^2 - dz^2.} \tag{1.23}$$

This is precisely the Rindler metric (1.4). Schwarzschild time near the horizon is a hyperbolic angle multiplied by $4GM$.

Question from the audience. Why may the radial term simply be replaced by $d\rho^2$, rather than replacing dr by $d\rho$? [a]

Response. The definition is $d\rho^2 = [r_s/(r - r_s)]dr^2$. It is the complete weighted radial term that equals $d\rho^2$; one never asserts $d\rho = dr$. The coordinate was chosen precisely to absorb that metric coefficient.

[a] Lecture timestamp: 01:03:38–01:05:35.

The coefficient

$$\kappa = \frac{1}{4GM} \tag{1.24}$$

is the Schwarzschild surface gravity in units $c = 1$. In the local form $ds^2 = (\kappa\rho)^2 dt^2 - d\rho^2 - \cdots$, a supported observer at proper distance ρ has leading acceleration $a \simeq 1/\rho$. Far from the horizon, Schwarzschild t again behaves as ordinary inertial time. Close to the horizon, $t/(4GM)$ behaves as

Rindler rapidity. The transition belongs to the coordinates; the local geometry at the horizon is regular.

1.5 The Horizon as Seen from Outside

The near-horizon reduction now tells us how an observer who remains outside describes infall. A stationary Schwarzschild observer is the black-hole analogue of a Rindler observer: staying at fixed r requires support, and the required acceleration grows without bound as $r \to r_s^+$. For that exterior system of clocks, the horizon lies at

$$\omega = \frac{t}{4GM} \longrightarrow +\infty.$$

Nothing reaches the horizon at a finite value of Schwarzschild time. This is not a statement about the infalling observer's proper time. It is a statement about the time coordinate tied to observers who refuse to fall.

Question from the audience. How does the infinite Rindler time translate into the black-hole picture? [a]

Response. The horizon is the $\omega = \infty$ boundary of the accelerated chart. A stationary observer can wait arbitrarily long, look back, and still assign the infalling body a position outside the horizon. Very close to the horizon this is exactly the Rindler description. Far away the metric returns to ordinary asymptotically flat coordinates, so the hyperbolic interpretation of $t/(4GM)$ is only local to the horizon.

[a]Lecture timestamp: 01:10:13–01:12:59.

How, then, can a black hole grow if the exterior observer never sees matter cross? In the exterior description, the infalling energy becomes concentrated ever closer to the old horizon. Its gravitational field changes the geometry, and the new horizon encloses it. A fully dynamical account requires solving Einstein's equations with infalling stress-energy; the fixed Schwarzschild metric alone cannot describe the growth.

Question from the audience. If nothing is seen to cross, how can the black hole become larger? [a]

Response. The incoming energy is deposited arbitrarily close to the horizon and changes the spacetime. The horizon grows outward and merges with that energy. Treating the matter as a test object in an eternally fixed metric misses precisely this backreaction.

[a]Lecture timestamp: 01:13:16–01:14:39.

The optical signature follows directly from the null rays. Suppose Alice carries an atomic dipole that emits one oscillation after another at a fixed frequency in her local frame. Successive crests reach Bob at increasingly separated values of his proper time. In the local Rindler picture this is an ordinary Doppler shift caused by their increasing relative rapidity; in Schwarzschild language it is the gravitational redshift. These are two descriptions of the same near-horizon effect.

Bob first receives optical light, then infrared, microwaves, and radio waves of progressively longer wavelength. He receives photons less frequently, and each photon has lower energy,

$$E_{\text{obs}} = \hbar\omega_{\text{obs}}, \qquad \omega_{\text{obs}} \longrightarrow 0. \tag{1.25}$$

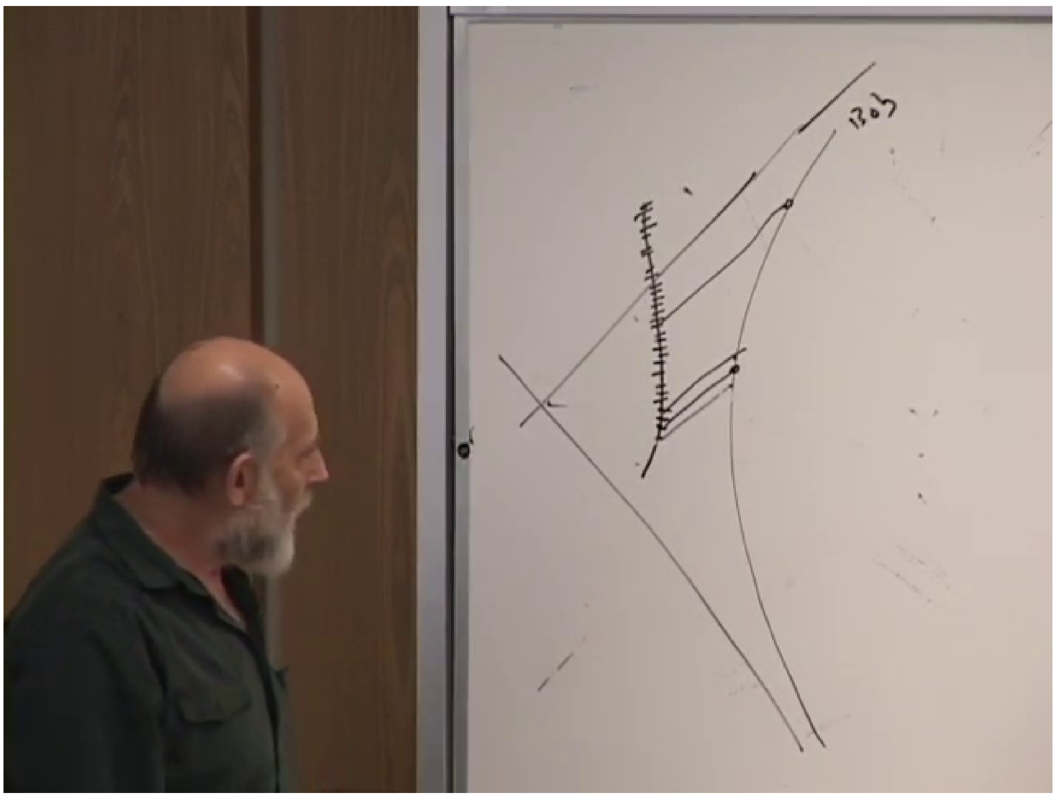

Figure 1.11: Successive wave crests from the infalling Alice arrive farther apart on Bob's supported worldline. The same construction is read as a Doppler shift in Rindler coordinates and as gravitational redshift in Schwarzschild coordinates.

Lecture frame: 01:17:00.

Every visible process on Alice appears to slow for the same reason: all information reaches Bob through these increasingly delayed signals.

Question from the audience. Why begin with optical light rather than radio signals? [a]

Response. Beginning at a short wavelength makes the redshift sequence visible in familiar bands: optical, infrared, microwave, and then successively longer radio waves. Starting with radio waves would leave no equally familiar names for the increasingly long wavelengths.

[a] Lecture timestamp: 01:14:53–01:15:24.

For an emitter with finite proper frequency, only finitely many oscillations fit into any finite interval of Alice's proper time. There is therefore a last crest emitted before crossing within that idealized pulse train, and Bob receives it only after an enormous delay and redshift. Increasing the source frequency inserts more crests, but does not change the limiting geometry.

Question from the audience. Is the longest possible wavelength necessarily of order the black-hole radius? [a]

Response. That conclusion does not follow from the classical redshift diagram alone. A finite-frequency source has a last crest before crossing, but classical theory permits arbitrarily low observed frequencies. A physical cutoff or a statement about one photon per mode requires quantum mechanics, so the proposed maximum-wavelength argument is set aside. Alice, meanwhile, sees Bob recede toward the speed of light and sees his processes Doppler slowed.

[a] Lecture timestamp: 01:18:58–01:20:29; 02:16:54–02:17:11.

1.6 Communication Across the Horizon

The horizon is not locally violent. Put Alice and Shirley into free fall side by side. They remain in ordinary local communication as they cross, provided their separation is small compared with the curvature scale. Neither sees the other freeze. The freezing belongs to Bob's supported exterior description, not to the freely falling laboratory.

Question from the audience. Is the mass really collapsed to a point in the exterior solution? [a]

Response. The Schwarzschild metric specifies the vacuum geometry outside a spherical source; it does not by itself say that ordinary matter has always occupied $r = 0$. For a star, the Schwarzschild exterior ends at the stellar surface. After black-hole formation, the same vacuum form covers the exterior while the infalling matter and the singular interior require the global collapse geometry.

[a] Lecture timestamp: 01:21:09–01:22:13.

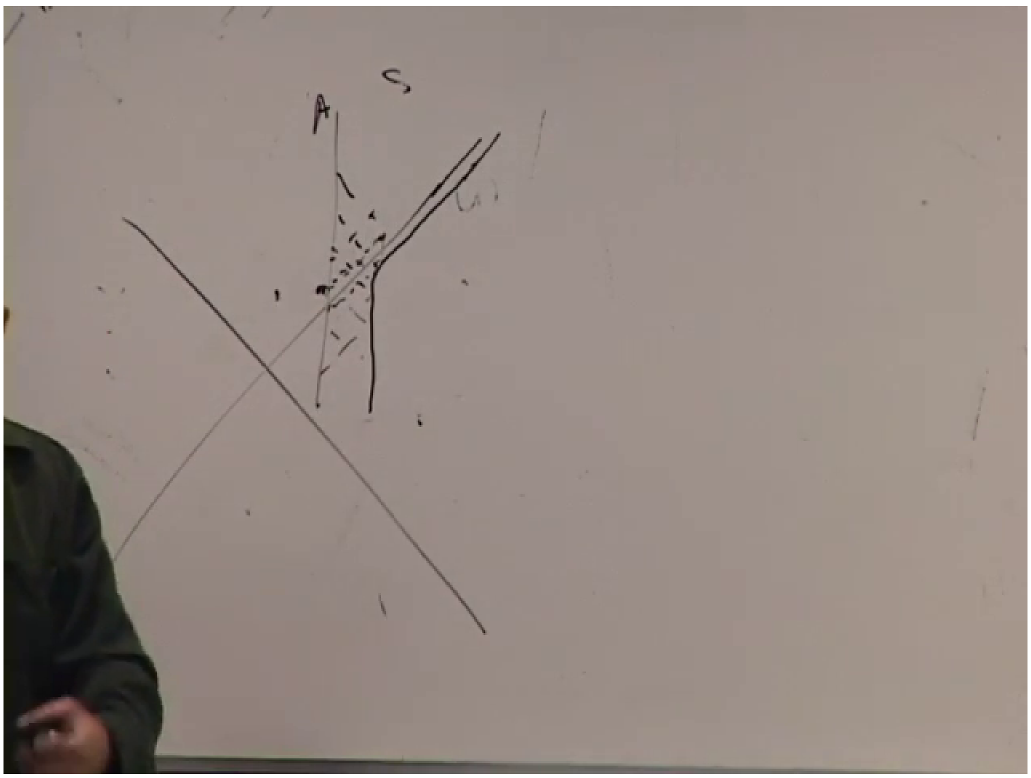

Figure 1.12: Two freely falling companions cross the horizon together. Their local exchange of signals remains regular even though a supported exterior observer receives Alice's signals ever more slowly.

Lecture frame: 01:24:50.

Question from the audience. If Alice and Shirley fall together, does either one see the other disappear or become ionized at the horizon? [a]

Response. No local event marks the crossing. Nearby freely falling observers continue to exchange light normally. A supported observer attributes extreme redshift and delay to Alice; Alice does not encounter a hot membrane in this classical discussion. Any claim about ionization involves the observer, the state of the radiation, and quantum physics, not a classical curvature singularity at the horizon.

[a] Lecture timestamp: 01:22:36–01:25:43.

Nor can Alice evade the interior merely by accelerating sideways. Near the horizon the transverse directions remain ordinary:

$$ds^2 \simeq (\kappa\rho)^2 dt^2 - d\rho^2 - dy^2 - dz^2. \tag{1.26}$$

At fixed ρ, a finite transverse velocity must still satisfy $(dy/dt)^2 + (dz/dt)^2 < (\kappa\rho)^2$. The allowed coordinate speed shrinks as $\rho \to 0$. A worldline that tries to skim the horizon at fixed r would need unbounded acceleration; once inside, fixed r is not a timelike option at all.

Question from the audience. Could an infalling observer accelerate tangentially and travel along the horizon instead of toward the singularity? [a]

Response. A massive observer must remain inside the local light cone. Exactly on the horizon the outward null direction is the horizon itself; a timelike observer cannot remain there. Inside, all future-directed timelike and null directions lead toward smaller r. Sideways motion changes where the observer arrives, not whether the future singularity is reached.

[a] Lecture timestamp: 01:27:09–01:31:54.

1.7 Inside the Horizon: Fixed Radius Becomes Spacelike

The apparent singularity at $r = r_\mathrm{s}$ disappears in a regular coordinate system, but the singularity at $r = 0$ does not. The Kretschmann scalar in (1.8) is finite at $r = 2GM$ and diverges at $r = 0$.

The reversal of coordinate roles inside the horizon is equally important. In the Schwarzschild metric, the factor $1 - r_\mathrm{s}/r$ is negative for $r < r_\mathrm{s}$. Consequently the r-direction contributes with the timelike sign while the t-direction contributes with a spacelike sign. A line of fixed r, which represented a supported observer outside, is spacelike inside. It cannot be the worldline of matter.

Calling $r = 0$ spacelike means that it is not a place waiting at the center in the ordinary Newtonian sense. Inside the black hole it is more like a future time: just as an exterior observer cannot avoid tomorrow, an interior observer cannot remain at fixed r and avoid the singularity. Every future-directed timelike path reaches it in a finite proper time.

Question from the audience. Does a freely falling observer experience any special event at the horizon, and how long remains before the singularity? [a]

Response. Nothing locally singular occurs at crossing; the strange behavior belongs to Schwarzschild coordinates. The proper time from the horizon to $r = 0$ is finite and is of order the black-hole radius in units $c = 1$, hence of order GM. The exact coefficient depends on the infalling geodesic.

[a] Lecture timestamp: 01:32:17–01:34:45.

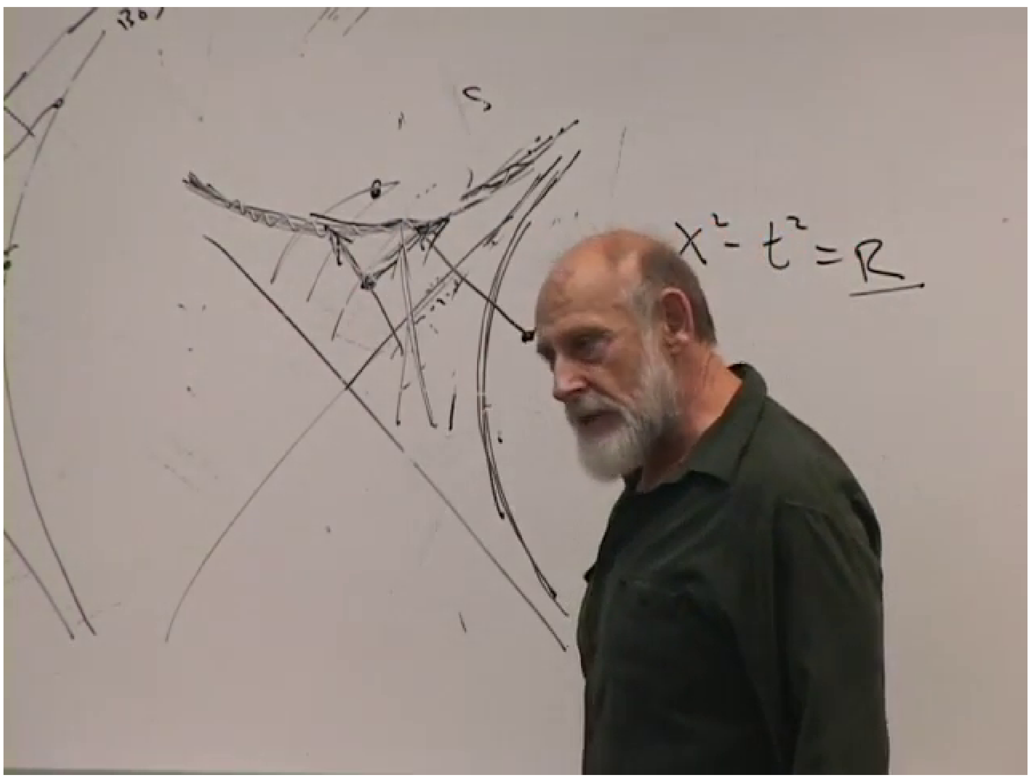

Figure 1.13: The maximally extended local picture: the horizon is a regular null boundary, whereas $r = 0$ is a spacelike future singularity reached by every future-directed interior worldline.

Lecture frame: 01:33:25.

The radial tidal scale is set by

$$|R_{\hat{0}\hat{r}\hat{0}\hat{r}}| \sim \frac{GM}{r^3}. \tag{1.27}$$

At the horizon this scales as

$$\frac{GM}{r_s^3} = \frac{1}{8G^2M^2}.$$

Thus a stellar-mass black hole can produce destructive horizon-scale tidal forces, whereas the horizon of a supermassive black hole can be locally gentle. In either case the tidal field diverges as $r \to 0$.

Question from the audience. Is the singularity an impact with matter, and does a larger black hole make the crossing safer? [a]

Response. The singularity is not an ordinary material surface that one strikes. It is the boundary where the classical curvature diverges and the theory ceases to predict. Increasing M reduces the curvature and tidal force at the horizon as M^{-2}, so a very large black hole may be crossed without immediate discomfort. The traveler still reaches the high-curvature region later in finite proper time.

[a] Lecture timestamp: 01:35:23–01:40:21.

1.8 Rods, Clocks, and Coordinate Roles

Coordinates are labels, not instruments. Outside the horizon, a static lattice of rods and supported clocks can give operational meaning to Schwarzschild r and t. That lattice cannot be continued through the horizon: the required support acceleration diverges there, and fixed r becomes spacelike inside. One may continue the coordinate symbols algebraically, but one must not continue their exterior measuring instructions unchanged.

Question from the audience. If r becomes timelike and t spacelike, do rods become clocks and clocks become rods? [a]

Response. No physical object changes its nature. The signs say that the coordinate directions have exchanged causal roles, not that a meter stick has turned into a clock. Inside, r cannot be measured by a chain of stationary rulers because no stationary observers exist there. A freely falling local laboratory keeps its ordinary rods, clocks, and Minkowski metric while it crosses.

[a] Lecture timestamp: 01:40:30–01:45:25.

1.9 Where the Exterior Metric Comes From

The Schwarzschild metric is not obtained by placing a point mass into the line element by hand. Begin with Einstein's equation,

$$G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = 8\pi G T_{\mu\nu}. \tag{1.28}$$

Outside a spherical distribution of matter, $T_{\mu\nu} = 0$, so

$$G_{\mu\nu} = 0. \tag{1.29}$$

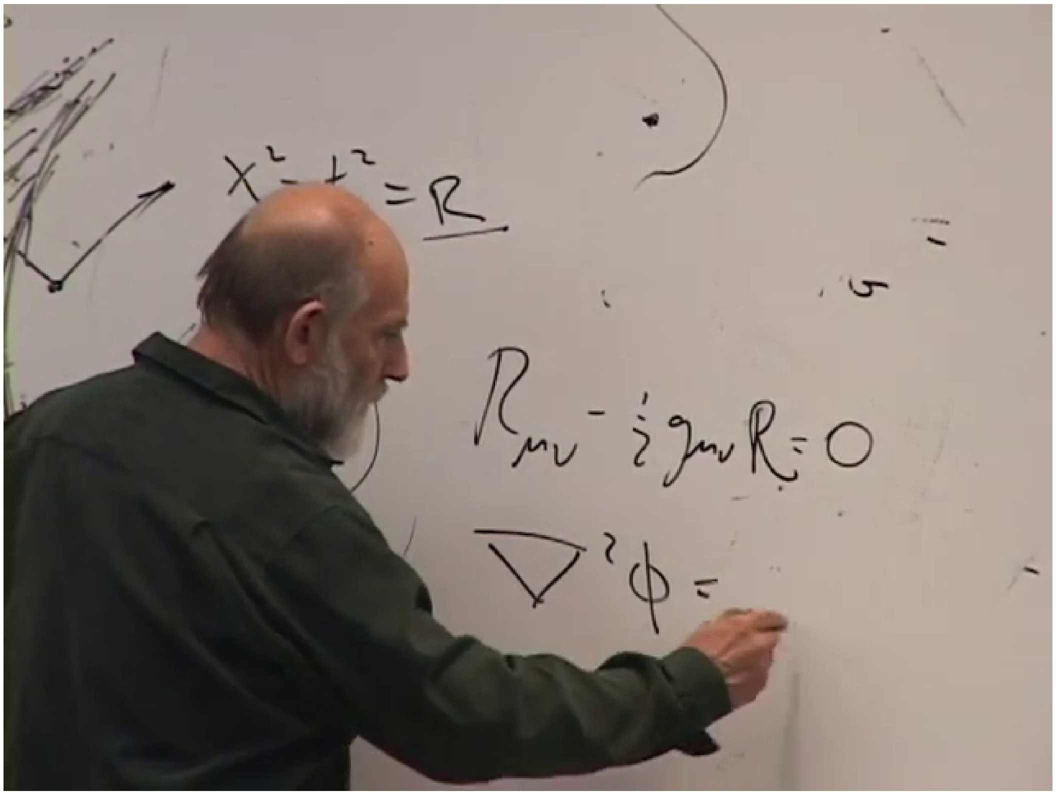

Figure 1.14: The vacuum Einstein equation and its Newtonian analogue. The exterior field is source-free locally, while its integration constant records the mass enclosed by the vacuum region.

Lecture frame: 01:47:55.

Spherical symmetry and asymptotic flatness then select the Schwarzschild family. Birkhoff's theorem adds the striking fact that the spherically symmetric vacuum exterior is static even when the source itself changes spherically with time. The integration constant is the total mass M enclosed by the vacuum region.

The Newtonian analogue makes the logic transparent:

$$\nabla^2\Phi = 4\pi G\rho_{\mathrm{m}}, \qquad \nabla^2\Phi = 0 \quad \text{outside the matter.} \tag{1.30}$$

For a spherically symmetric potential that vanishes at infinity, the exterior solution is

$$\Phi(r) = -\frac{GM}{r}. \tag{1.31}$$

The differential equation is homogeneous in the exterior, yet the boundary condition carries the source mass. General relativity works in the same structural way, although its field equations are nonlinear and the unknown is the metric rather than a scalar potential.

Question from the audience. What matter distribution was assumed when the Schwarzschild metric was derived? [a]

Response. No detailed interior distribution is needed for the vacuum exterior. One assumes spherical symmetry, solves $G_{\mu\nu} = 0$ outside a sphere containing all the matter, and requires asymptotic flatness. The enclosed mass fixes the one remaining constant, just as it fixes the coefficient $-GM$ in the Newtonian solution $\Phi = -GM/r$.

[a] Lecture timestamp: 01:46:19–01:50:24.

Shrinking the enclosing sphere in Newtonian theory leaves a source-free solution everywhere except at the point source. In a collapsed Schwarzschild geometry, however, saying that the matter is at $r = 0$ must be interpreted causally: $r = 0$ is the spacelike future singularity, not a persistent point in space on which matter sits while the exterior evolves.

Question from the audience. Where is the material after it falls into the black hole? [a]

Response. If one insists on locating it in the ideal Schwarzschild solution, its future ends at $r = 0$. Everywhere away from the collapsing matter and the singular boundary, the vacuum equation holds. The analogy with a Newtonian point source is useful for the exterior field, but it fails geometrically because the relativistic $r = 0$ is spacelike rather than a timelike point at the center.

[a] Lecture timestamp: 01:50:31–01:52:26.

1.10 Why Collapse Is Not Newtonian Point Aiming

In Newtonian many-body mechanics, collapse to an exact point is exceptional. The slightest angular momentum generally makes particles miss one another and pass through or orbit. Even if a point singularity were postulated, an additional particle would hit it only under exquisitely aimed initial conditions. This intuition once encouraged the belief that generic gravitational systems could not form singularities.

The event horizon changes the question. Once a future-directed timelike trajectory crosses it, no choice of aim or angular momentum returns that trajectory to the exterior. The singularity is not a spatial target that can be missed; it is a future spacelike boundary. Newtonian point-aiming arguments therefore do not prevent black-hole formation.

Question from the audience. Is the same angular momentum that stabilizes a cluster enough to prevent relativistic collapse? [a]

Response. Angular momentum makes exact Newtonian point collapse nongeneric and helps long-lived systems such as clusters avoid immediate collapse. But if a configuration lies inside its event horizon, every future-directed causal path remains inside and heads toward the singular future. The old Newtonian argument treated the singularity as a point one could miss and did not account for the horizon.

[a] Lecture timestamp: 01:52:29–01:56:31.

Pressure can resist collapse before a horizon forms. Classical Hamiltonian evolution preserves phase-space volume by Liouville's theorem. Quantum statistics and the uncertainty principle introduce additional scales and, for fermions, degeneracy pressure. These are real sources of support, but they do not imply that an arbitrarily massive object must remain outside its Schwarzschild radius.

Question from the audience. Is phase-space incompressibility a consequence of Planck's constant, and can it always stop collapse? [a]

Response. Liouville incompressibility is already a classical theorem; it is not caused by $\hbar$. Quantum mechanics makes a phase-space cell physically significant and can generate pressure, but pressure is a local material response. A sufficiently large mass can have a Schwarzschild radius at a very low mean density, long before microscopic compression becomes extreme.

[a] Lecture timestamp: 01:56:33–01:58:22.

1.11 Mean Density and Horizon Scale

For a body of radius R and mean density $\bar{\rho}$,

$$M = \frac{4\pi}{3}\bar{\rho}R^3, \qquad R = \left(\frac{3M}{4\pi\bar{\rho}}\right)^{1/3}. \tag{1.32}$$

At fixed $\bar{\rho}$, the material radius grows as $M^{1/3}$, while the Schwarzschild radius grows as M. No matter how small the density is, a sufficiently large mass satisfies

$$R < 2GM. \tag{1.33}$$

At the threshold $R = 2GM$, the corresponding mean density is

$$\bar{\rho}_{\mathrm{BH}} = \frac{3M}{4\pi(2GM)^3} = \frac{3}{32\pi G^3 M^2}. \tag{1.34}$$

The larger the black hole, the smaller its horizon-scale mean density.

This scaling explains why black-hole formation does not require matter to begin at a universal enormous density. For stellar masses, material support and high density are central to the collapse history. For sufficiently large M, the horizon condition itself can occur at a modest average density.

1.11.1 The cosmological comparison

For approximately uniform vacuum energy density, the natural horizon scale is cosmological. In a spatially flat de Sitter approximation,

$$H^2 = \frac{8\pi G}{3}\rho_{\mathrm{vac}}, \qquad R_{\mathrm{dS}} = H^{-1}. \tag{1.35}$$

The Schwarzschild radius associated with the vacuum energy enclosed inside a sphere becomes comparable to the sphere's radius precisely at this scale. The analogy is useful but must not be mistaken for identity: a de Sitter observer is surrounded by a cosmological horizon, not trapped inside a Schwarzschild black hole.

Question from the audience. Does the vacuum energy of our universe give a Schwarzschild radius, and does that mean we live inside a black hole? [a]

Response. The associated scale is the cosmological horizon, of order the inverse Hubble parameter for vacuum-dominated expansion. It is an inside-out analogue: distant objects recede and pass beyond our causal horizon, whereas inside a black hole all future directions lead inward toward a singularity. A cosmological horizon is not the interior of a Schwarzschild black hole.

[a] Lecture timestamp: 02:02:26–02:04:24.

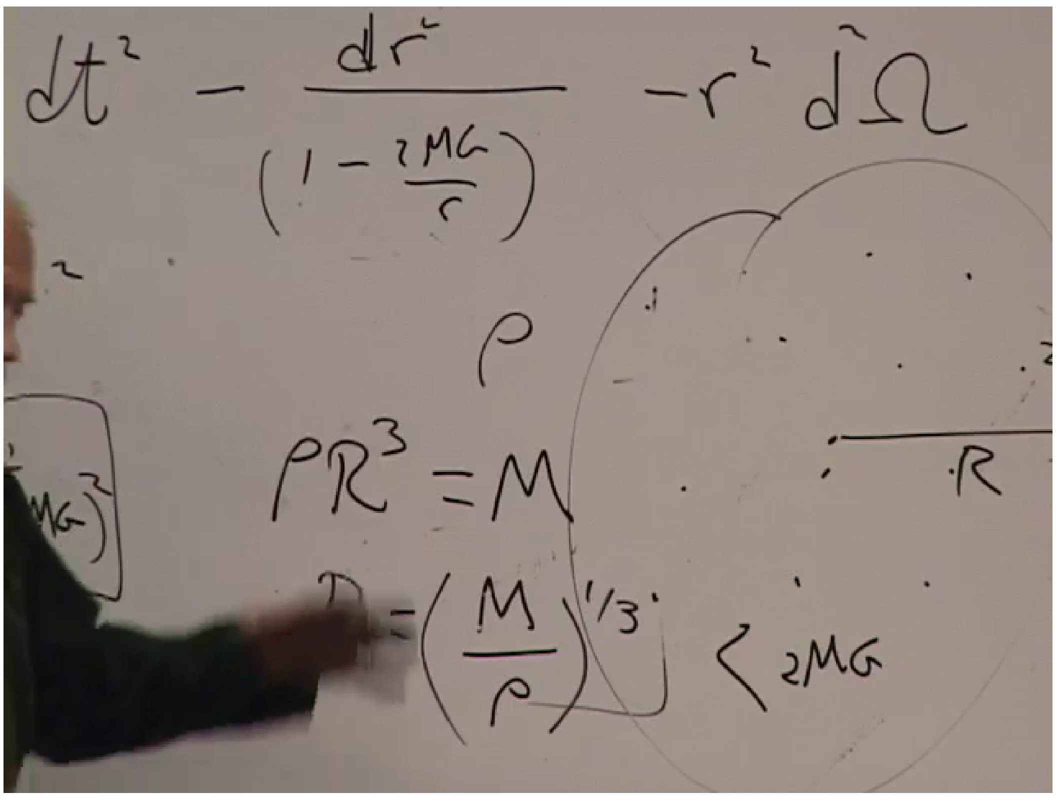

Figure 1.15: For fixed mean density, $R \sim (M/\bar{\rho})^{1/3}$, whereas $r_s = 2GM$ grows linearly with mass. A sufficiently large region therefore lies within its Schwarzschild radius even when the material is rarefied.

Lecture frame: 02:02:10.

1.12 What Classical Theory Cannot Continue

The phrase end of time is shorthand, not a claim that the traveler's clock takes infinitely long to arrive. The proper time to $r = 0$ is finite. What ends is the classical spacetime predicted by the Schwarzschild solution. As the curvature, density, and tidal gradients grow without bound, any fixed body of known matter is driven beyond the domain where its familiar description applies.

Question from the audience. If the singularity is the end of time, how can an observer actually reach it? [a]

Response. The observer's proper time to the singularity is finite. Calling it an end means that the classical solution has no regular future continuation there. It is not meaningful to imagine the observer stopping at fixed $r = 0$; fixed r was already spacelike inside. Curvature and material scales run beyond every regime covered by the classical theory.

[a] Lecture timestamp: 02:04:25–02:06:40.

There is also an operational way to see the loss of ordinary structure. Two neighboring constituents can influence one another only if a light signal has time to pass between them. As the singularity approaches, the remaining proper time becomes shorter than the required communication time over successively smaller separations. First separated atoms lose causal contact, then electrons and nuclei, and eventually nuclear constituents. The classical picture is not a final microscopic theory; it says that no finite bound structure can preserve its internal communication all the way to the singular boundary.

Question from the audience. Is the singularity merely irrelevant because no signal from it can return to the exterior? [a]

Response. It is causally irrelevant to an observer who remains outside, but an infalling observer cannot turn it into an experiment with a reported result. Near the singularity, separated parts of the apparatus lose causal contact before they could establish or transmit a shared observation.

[a] Lecture timestamp: 02:06:40–02:08:53.

The Rindler approximation cannot be used to infer this behavior. It was obtained under $|r - r_s| \ll r_s$ and is valid only near the regular horizon. The global interior diagram and the curvature divergence come from the full Schwarzschild geometry.

Question from the audience. Does the flat Rindler approximation remain valid when we approach $r = 0$? [a]

Response. No. The approximation was made by expanding about $r = r_s$. Moving toward $r = 0$ leaves that regime. The regularity of the horizon is established locally; the spacelike singularity requires the full solution and has genuine curvature.

[a] Lecture timestamp: 02:08:53–02:09:43.

1.12.1 There Is No Classical Other Side of the Singularity

At the horizon, a bad coordinate can be replaced by a regular one and the metric extends smoothly. At $r = 0$, the invariant curvature diverges. Substituting a negative number for the areal radius in the Schwarzschild formula is therefore not an analytic continuation through a regular surface. Classical general relativity supplies no spacetime beyond that singular boundary.

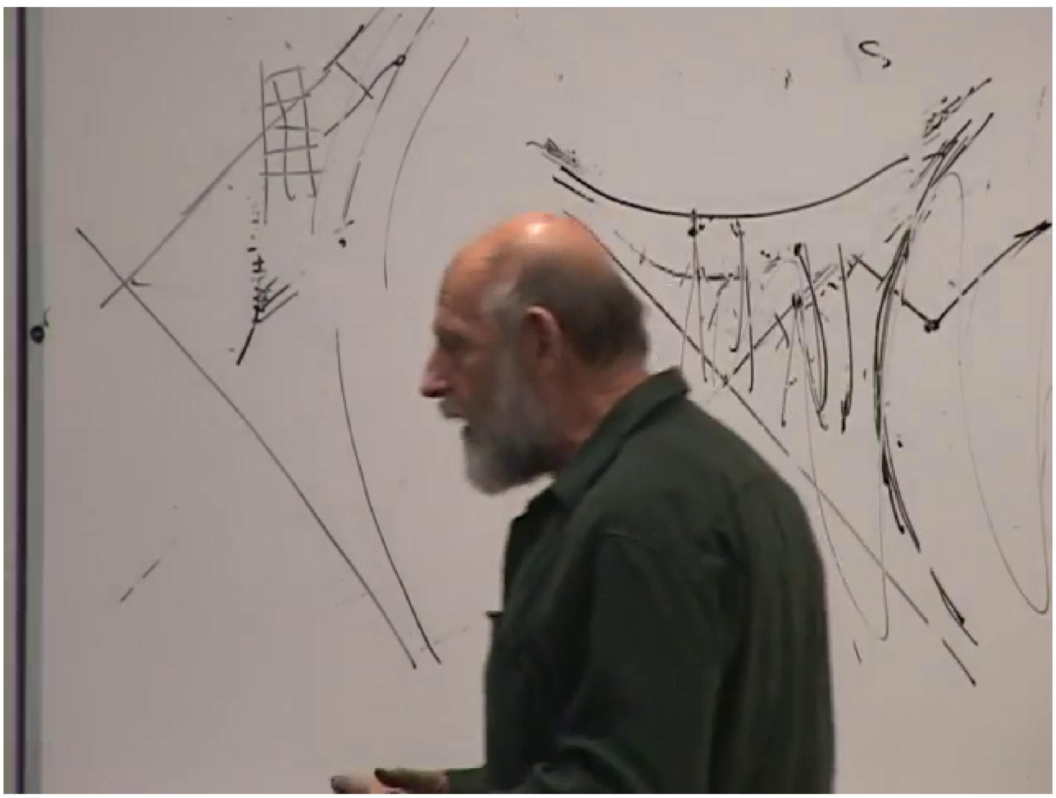

Figure 1.16: Near the spacelike singularity, neighboring worldlines have too little remaining proper time to exchange signals. Bound structures lose causal communication on successively smaller scales.

Lecture frame: 02:08:45.

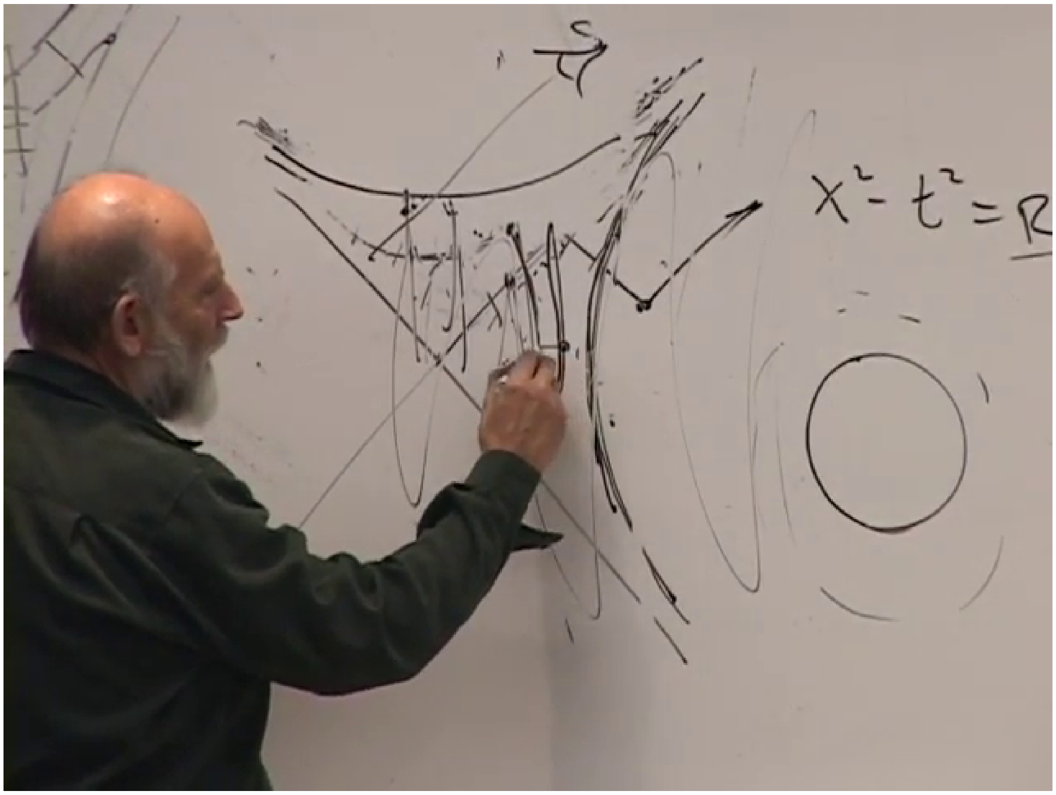

Figure 1.17: Alice and Shirley carry a finite meter stick through the horizon. Their local apparatus remains regular; the dramatic exchange belongs to the Schwarzschild coordinate labels.

Lecture frame: 02:13:00.

Question from the audience. Can a mathematician continue the metric to negative r and call that the other side of the singularity? [a]

Response. One may manipulate a formula beyond its physical domain, but the curvature singularity blocks a regular spacetime extension. Unlike the horizon, there is no coordinate change that removes the divergence and no observer can cross, return, and test a putative continuation. Questions about information and quantum resolution require physics beyond this classical lecture.

[a] Lecture timestamp: 02:09:48–02:12:34.

1.13 Operational Tests at the Horizon

Carry a meter stick across the horizon, with Alice at one end and Shirley at the other. Provided the stick is small compared with the curvature scale, it remains an ordinary local object. It does not turn into a clock, and Alice's clock does not turn into a ruler. Only the Schwarzschild coordinate directions exchange causal roles.

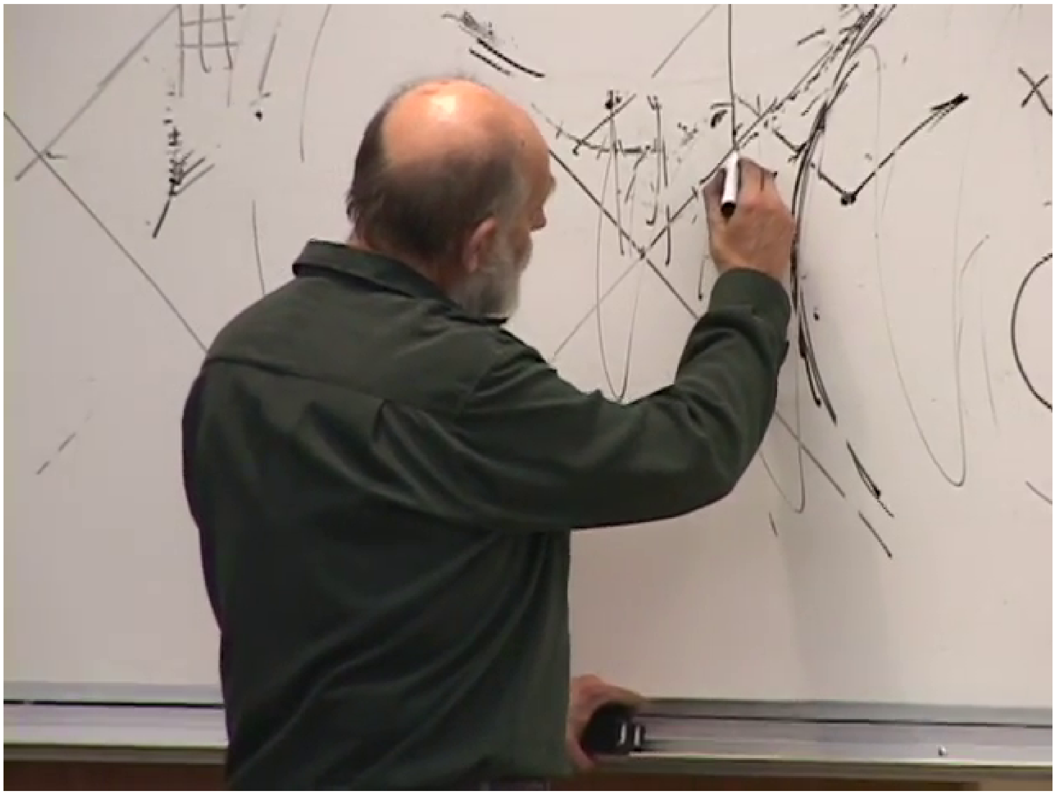

Figure 1.18: Alice sends pulses to Bob's reflector. Successive return signals are delayed and redshifted as the horizon is approached, but no local discontinuity appears at crossing.

Lecture frame: 02:16:20.

Question from the audience. Does Alice's meter stick become a clock when it crosses the horizon? [a]

Response. Absolutely not. The stick and its endpoints pass through normally in a freely falling frame. The symbol r no longer labels a spacelike measuring direction inside, but naming a coordinate r does not require it to be measured by rulers in every region. Local rods and clocks retain their ordinary roles.

[a]Lecture timestamp: 02:12:40–02:14:24.

A second experiment uses reflected light. Before crossing, Alice sends a pulse outward toward Bob's reflector and receives the return. As the emission point approaches the horizon, both legs of the trip take longer in the exterior/Rindler description. Motion of the reflector relative to Alice redshifts the return. Alice observes progressively delayed and redshifted echoes, not a local image or flash that announces the horizon at the instant of crossing.

Question from the audience. Can Alice locate the horizon by sending a signal to Bob's reflector and watching the echo? [a]

Response. She sees the round-trip time grow and the reflected photons become strongly redshifted because Bob is accelerating away in the local Rindler picture. These changes are continuous. No return signal received at the crossing event identifies a local horizon, and after crossing no future-directed signal reaches a reflector that remains outside.

[a]Lecture timestamp: 02:14:26–02:16:52.

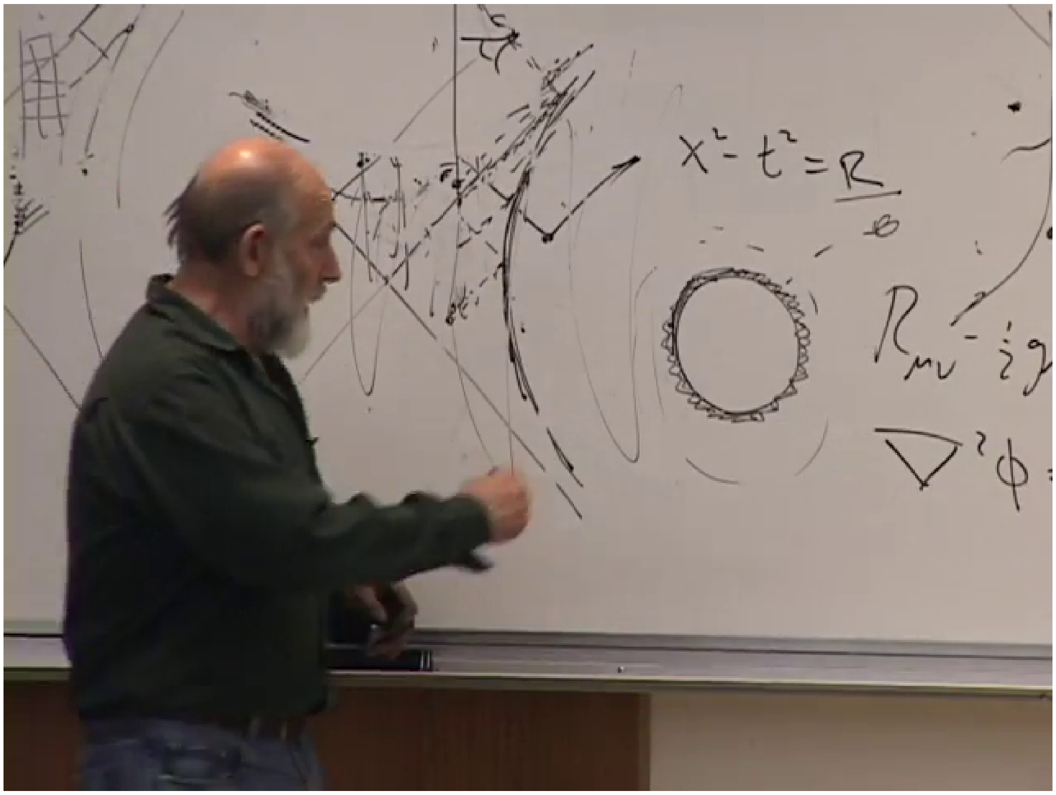

Figure 1.19: The exterior slicing represents infalling material as increasingly thin layers near the horizon. The surrounding field records the enclosed mass even though no signal escapes from the interior.

Lecture frame: 02:18:45.

1.14 The Exterior Thin-Shell Description

From the viewpoint of a Schwarzschild observer, infalling layers approach the horizon asymptotically and become ever thinner and more redshifted. It is therefore permissible within that exterior description to picture the accumulated mass as sedimentary layers concentrated just outside the horizon. This is a description tied to the exterior time slicing, not a material membrane encountered by a freely falling observer.

Question from the audience. If gravitons cannot escape the black hole, how can the exterior gravitational field persist? [a]

Response. The picture of force as particles shuttling back and forth should not be taken literally. Classically, the exterior geometry is the solution of the field equations with the conserved mass as boundary data. If one insists on an exchange-particle heuristic, the exterior slicing places the effective source in the ever-thinner layers just above the horizon; no graviton must carry news outward from the singular interior.

[a] Lecture timestamp: 02:17:13–02:18:58.

1.15 Rotating Black Holes

Astrophysical black holes generally rotate. Their stationary exterior is described by the Kerr solution rather than the Schwarzschild solution. In units $c = 1$, a Kerr horizon exists only when

$$|J| \leq GM^2, \qquad a_* = \frac{J}{GM^2}, \qquad |a_*| \leq 1. \tag{1.36}$$

Rotation flattens the horizon and introduces frame dragging, but the dimensionless spin is bounded. A black hole therefore cannot become an arbitrarily flattened material disk.

Question from the audience. Could a rotating black hole contain a stable binary system orbiting inside it, and would gravitons from that system have to escape? [a]

Response. No stationary internal binary can remain hidden inside the horizon and keep orbiting forever. Once inside, its future-directed motion belongs to the black-hole interior and ends at the singular region. The exterior Kerr field is characterized by global charges such as M and J; it does not require signals or gravitons to escape from an enduring mechanical system inside.

[a] Lecture timestamp: 02:19:00–02:20:22.

1.16 Summary

The Schwarzschild horizon is locally simple. Proper distance from the horizon and the rescaled time $t/(4GM)$ turn its metric into the Rindler metric of flat spacetime in accelerated coordinates. This explains both sides of the familiar account: a freely falling observer crosses in finite proper time without encountering local singular curvature, while a supported exterior observer receives signals that are delayed and redshifted without bound.

The interior is not merely another Rindler wedge. Fixed r becomes spacelike, the singularity at $r = 0$ is a spacelike future boundary, and the curvature invariant grows as r^{-6}. Angular momentum, pressure, and phase-space constraints can delay or prevent a system from entering its Schwarzschild radius, but after a horizon forms they do not provide an outward future. The vacuum field equations determine the exterior from symmetry and conserved mass, the horizon-scale mean density decreases as M^{-2}, and the classical description finally ends where causal communication and curvature scales run beyond its domain. Rotating black holes modify the geometry but preserve the central lesson: a horizon is regular locally and decisive globally.